

Nicholas Henry

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EDUCATION

Northeastern University Boston, MA
Bachelor of Science in Mechanical Engineering and Computer Science May 2029
Coursework: Diff Eq, Lin Alg, Discrete Structures, Program Design and Implementation, Statics **GPA: 3.79**
Activities: Northeastern Electric Racing (FSAE), Students for the Exploration and Development of Space, Deans List
Studied a Semester Abroad at Queen's University Belfast in Northern Ireland

SKILLS

Languages: Python, C, Java, C#, JavaScript, SQL, HTML/CSS, TypeScript, C++
Frameworks & Libraries: React, Next.js, Node.js, Tensorflow, NumPy, Flask, Dash, Tailwind
Tools & Platforms: Git, Docker, Github Actions, Redis, Supabase, Linux, ROS 2, PostgreSQL, CI/CD, Claude Code, Raspberry Pi, Kubernetes
Engineering Skills: SolidWorks, MATLAB, Project Management, Problem Solving, Cross-Functional Collaboration
Certifications & Training: IBM Artificial Intelligence Fundamentals, Microsoft C# Certification, GitHub Foundations

EXPERIENCE

Northeastern Electric Racing - Python, Dash, React, Docker, Git, Redis, Grafana Boston, MA
Lead Software Engineer for Performance Analysis Team Jan 2026 – Present

- Develop a parametric multi-body vehicle simulation tool for mechanical engineers to test and examine how changes to geometry and other variables affect the speed and times of the car on custom created tracks
- Utilize Python for the backend to do all the physics with Plotly Dash, and React for the frontend of the web app to create all the graphs and metrics making the UI easy for engineers to use effectively
- Conduct meetings, delegate work to members, perform code reviews, and research how to improve the physics engine

Students for the Exploration and Development of Space - Python, ROS 2, Git, Linux, Docker Boston, MA
Software Engineer for Network Exploration Swarm Toolkit Team Jan 2026 – Present

- Program for a team to create a swarm of C.O.B.R.A. (Crater Observing Bio-inspired Rolling Articulator) snake robots to explore and develop multi-agent collaboration in order to complete teamwork based exploration tasks
- Create the camera sensor ecosystem consisting of a simultaneous mapping program for multi-agent systems

SEONeeded - Wordpress, SEOMoz, Semrush Miami, FL
Web Optimization Engineer Oct 2025 – Aug 2026

- Performed client-facing tasks like identifying keywords, website maintenance, and strategies to target in order to drive more traffic for certain services increasing click-on rate and sales for 15 clients across the Greater South Florida region
- Adapted to the new wave of generative engine optimization space by learning what makes LLM models reference client's websites and how certain worded prompts get tailored responses, recommendations, and links to sources

Search Engine Optimization Intern Jun 2025 – Sep 2025

- Drove a 54% increase in website traffic through targeted optimization by leveraging WordPress and keyword research

PROJECTS

SheepFinder - React, Next.js, Vercel, SQL, Git, Tailwind May 2026 - Present

- Developed a Next.js and React web app based on time studying abroad in Northern Ireland designed to aid farmers and create an ecosystem where townspeople can report sightings of lost animals
- Held a focus group with farm owners to gain an insight into what the main issue is and the best ways to solve and build an app around that feedback like how sheep markings are designated across farms in different towns
- Connected 10+ unique animals with their farmers and heard from a farm of the financial benefits of the app

Portfolio Website - HTML/CSS, JavaScript, SQL, Supabase, Github Actions, Git Mar 2025 - Present

- Built a site with HTML/CSS and Javascript deployed through Github Pages with a custom domain to display projects
- Applied SQL databases with Supabase to have a global comment section where visitors can write something
- Use Github Actions in order to autoupdate my resume on the site to match this document as it changes in Google Docs

Text Transcriber - Python, TensorFlow, Google APIs, Git Jul 2025

- Designed and trained an AI model using TensorFlow on 100,000+ character images to recognize and convert handwriting into text and to then convert into audio with Google TTS as part of a 3-person team
- Tested writing and signage with a group of visually impaired participants to reduce caretaker dependency; results from the limited cohort were an 88% rate of correctly understanding the written text which greatly improved independence